

Scaling Portfolio-Level Reinforcement Learning via Anchor-Token Cross-Attention

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Abstract

TradingGame is a rule-constrained portfolio-management reinforcement learning environment: an actor-critic policy chooses buy/sell/hold actions jointly across a candidate universe of N stocks under a shared cash constraint, trained with Proximal Policy Optimization (PPO). The current policy network aggregates cross-candidate context with full self-attention over all N candidates plus a portfolio token, an $O(N^2)$ operation. This note (i) formalizes the environment and the existing architecture, (ii) reports a direct empirical characterization of that architecture’s GPU memory footprint — thirteen measurements on real hardware, fitted to a closed-form model with $R^2 = 0.9999999$ — showing the design cannot reach a $N=1,500$ -company universe at useful batch sizes, and (iii) proposes an $O(N(K+2))$ replacement, *anchor-token cross-attention*, in which each candidate attends only to a small fixed set of K real anchor stocks plus separate macro and portfolio tokens, rather than to every other candidate or to a single pooled summary. We show the redesign is necessary but not sufficient: removing the quadratic term exposes a previously-dominated linear term, whose coefficient we also measure directly, giving a closed-form bound on the batch size the redesign alone can reach at a given N .

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1 Introduction

`TradingGame` trains a single reinforcement-learning policy to manage a simulated portfolio of Indian (NSE/BSE) equities under an explicit rule set: a fixed starting cash balance; a 0.5% fee on every transaction; sale proceeds that settle into spendable cash only after a two-trading-day delay; a one-trading-day lock-up before a newly-bought lot can be voluntarily sold again; and an unconditional forced exit after ten trading days. The policy observes a capped *candidate universe* of N stocks at each decision step and must choose, jointly, which to hold, sell, or buy — subject to a single shared cash pool, so the decisions are coupled across candidates, not independent per-stock classifications.

The current implementation (`ActorCriticPolicy` in `packages/TradingGame/src/policy.jl`) caps N at 60. Growing the candidate universe toward index scale ($N \approx 1,500$) is desirable for realistic diversification, and is newly plausible now that training runs on a dedicated GPU rather than CPU alone. Section 4 shows why the current architecture cannot simply be pointed at a larger N : its cross-candidate attention mechanism is quadratic in N . Section 5 characterizes that cost directly, on real hardware, rather than by asymptotic argument alone. Section 6 derives a linear-cost replacement, and Section 7 shows why that replacement is a necessary but incomplete fix.

2 Related Work

Self-attention and its cost. Scaled dot-product self-attention [Vaswani et al., 2017] lets every element of a set condition on every other element, at $O(n^2)$ time and memory in the set size n . This is the mechanism the current `ActorCriticPolicy` uses across candidates.

Reducing attention to sub-quadratic cost. Several strategies replace all-pairs attention with attention against a smaller, fixed reference set. Lee et al. [2019] introduce the *Induced Set Attention Block* (ISAB): rather than n elements attending to all n others, they attend to $m \ll n$ learned *inducing points*, reducing cost from $O(n^2)$ to $O(nm)$ while remaining a permutation-invariant function of the input set — exactly the structure we adopt in Section 6, with the inducing points replaced by a small set of *real, interpretable* anchor stocks. Other lines of work achieve sub-quadratic attention via low-rank projection of the sequence dimension [Wang et al., 2020] or kernel approximations of the softmax [Choromanski et al., 2021]; we do not adopt these here because they trade the interpretability of "which stocks is candidate i attending to" for a compressed or randomized basis, which is a poor fit for a financial decision system where that interpretability is itself valuable.

Recurrent per-element encoders. Each candidate’s own price history is encoded independently with a Gated Recurrent Unit [Cho et al., 2014] before any cross-candidate mixing; this per-candidate encoding is where the dominant *linear-in- N* cost identified in Section 7 originates, and it is unaffected by whichever cross-candidate attention pattern is used downstream.

Combined policy/value networks trained by self-play or policy-gradient RL. AlphaGo Zero and its generalization [Silver et al., 2017, 2018] share a trunk between a policy head and a value head, trained end-to-end — the same actor-critic head structure `ActorCriticPolicy` uses, though those systems operate over a fixed-size board rather than a variable, rule-constrained financial action space. We train with Proximal Policy Optimization [Schulman et al., 2017] and Generalized Advantage Estimation [Schulman et al., 2016], in the tradition of actor-critic methods such as A3C [Mnih et al., 2016]; optimization uses Adam [Kingma and Ba, 2015]. LeCun et al. [2015] give a general account of the representation learning this architecture relies on throughout.

Internal precedent. This repository’s own `StockSwingPredictor` package made the analogous choice once before, for a different model: its first architecture (`DualCNN_v1`) aggregated a target stock’s peer context by *mean-pooling* up to 150 peer embeddings; the second (`DualCNN_v2`) replaced that pooling with cross-attention, specifically because a flat average cannot express *which* peers matter for a given target. Section 6 applies the identical argument to `TradingGame`: an earlier draft of this proposal itself made the mean-pool mistake it is meant to avoid, by aggregating all N candidates into one pooled "market pulse" token before realizing this reintroduced exactly the selectivity loss that motivated `DualCNN_v2` in the first place.

3 Environment and Decision Problem

We formalize the rules enforced by `packages/TradingGame/src/env.jl` (`TradingGameRules.txt`, repo root) as a discrete-time Markov decision process.

Portfolio state. At trading-hour index t , the portfolio holds spendable cash C_t , a set of reserved-cash lots $\mathcal{R}_t = \{(\rho_k, \tau_k)\}$ (amount, maturity trading-day index), and a set of open lots $\mathcal{L}_t = \{(i_\ell, q_\ell, p_\ell, d_\ell)\}$ (stock index, quantity, entry price, entry trading-day index). Portfolio value is

$$V_t = C_t + \sum_{(\rho, \tau) \in \mathcal{R}_t} \rho + \sum_{(i, q, p, d) \in \mathcal{L}_t} q \cdot P_{i,t}, \quad (1)$$

where $P_{i,t}$ is stock i ’s price at t — rules 2 and 6 of `TradingGameRules.txt`, restated identically.

Settlement, lock-up, forced exit. Let $\delta(t)$ be the trading-day index of hour t . A reserved-cash lot (ρ, τ) matures — moves into C_t — once $\delta(t) \geq \tau$, with $\tau = \delta(t_{\text{sell}}) + \tau_{\text{settle}}$, $\tau_{\text{settle}} = 2$ (rule 4). A lot (i, q, p, d) may be voluntarily sold only once $\delta(t) - d \geq \tau_{\text{lock}}$, $\tau_{\text{lock}} = 1$ (rule 9), and is *unconditionally* liquidated once $\delta(t) - d \geq \tau_{\text{max}}$, $\tau_{\text{max}} = 10$ (rule 8), independent of the policy’s action.

Action space. At each decision step the policy emits, jointly for every candidate $i \in \{1, \dots, N\}$, a discrete type and a continuous weight:

$$a_t = (a_{i,t}^{\text{type}}, w_{i,t})_{i=1}^N, \quad a_{i,t}^{\text{type}} \in \{\text{HOLD}, \text{SELL}, \text{BUY}\}, \quad w_{i,t} \in [0, 1]. \quad (2)$$

$w_{i,t}$ is only meaningful when $a_{i,t}^{\text{type}} = \text{BUY}$: it is the requested share of available cash, renormalized across every candidate selected BUY this step so that total notional-plus-fee never exceeds C_t (rule 5) — a deterministic masking layer (`resolve_actions` in `action.jl`), not something the policy must learn unaided. Every executed transaction pays a fee $\varphi = 0.005$ of its notional value (rule 10), and a SELL is only ever offered to the policy for lots that have cleared the lock-up in Eq. (2)’s constraint.

Reward. Consistent with rule 11's "continuously maximize portfolio value,"

$$r_t = \log \frac{V_t}{V_{t-1}}, \quad (3)$$

computed every hourly bar (not only on decision bars), so unrealized mark-to-market movement is credited even when the policy takes no action. Because Eq. (1) already carries reserved cash at full value throughout its settlement delay, a sale's only effect on r_t is the fee φ — the reward requires no separate transaction-cost penalty term.

4 Baseline Architecture: Full Self-Attention

`ActorCriticPolicy` (`policy.jl`, current implementation) computes an embedding $h_i \in \mathbb{R}^d$ for each candidate independently with a weight-shared GRU over its own $T=120$ -bar, $C=2$ -channel hourly window (normalised close and previous-day range), fuses in per-candidate news and holding-state features, and forms one additional *portfolio/macro token* m from the macro time series and the global portfolio scalars. The $N+1$ embeddings $\{h_1, \dots, h_N, m\}$ are then passed through full multi-head self-attention [Vaswani et al., 2017]:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad Q = K = V = [h_1; \dots; h_N; m] \in \mathbb{R}^{(N+1) \times d}. \quad (4)$$

Every one of the $N+1$ tokens is both a query and a key: the score matrix $QK^\top$ has $(N+1)^2$ entries, computed and differentiated once per attention head and once per batch element. This is the mechanism by which the network lets candidate i 's decision condition on candidate j 's current state — necessary for a *joint*, cash-constrained decision — but its cost scales as

$$\text{cost}_{\text{attn}}(N) = \Theta((N+1)^2). \quad (5)$$

Figure 1(a) shows this connectivity for a small illustrative N : every node connects to every other.

Game v2. The v2 rule set (`-game-version 2`) adds a *cash token*: cash is treated as a pseudo-stock and embedded from four scalars (cash and reserved-cash shares of portfolio value, utilisation of the cash cap, and time spent above it), giving $N+2$ tokens and $\Theta((N+2)^2)$ attention cost. The asymptotics and the conclusions below are unchanged. The GPU measurements in Section 5 were taken on the earlier $N+1$ -token, $C=3$ policy; the policy now has one fewer input channel and (under v2) one extra token, which shifts the fitted coefficients slightly but not the N^2 scaling. They have not been re-run.

5 Empirical Characterization of the Baseline

Asymptotic complexity alone does not say where a design stops fitting a particular GPU, so Eq. (5) was measured directly rather than taken on faith. For each (N, B) pair, `ActorCriticPolicy` was constructed, moved to GPU, and run through one full forward pass, backward pass, and Adam optimizer step [Kingma and Ba, 2015] on random input of the correct shape; peak resident GPU memory was read from `CUDA.used_memory()` immediately after. All measurements are on a single NVIDIA GeForce RTX 5060 Ti (16.72 GB total, ≈ 15.57 GB usable after driver/OS reservation).

Table 1: Measured peak GPU memory, ActorCriticPolicy forward + backward + Adam step.

N	Batch B	Peak memory (GB)	Result
60	4	0.122	fits
60	8	0.250	fits
60	16	0.493	fits
60	32	0.986	fits
60	64	1.971	fits
60	128	3.941	fits
60	256	7.882	fits
500	4	1.083	fits
500	8	2.165	fits
500	16	4.329	fits
500	32	8.657	fits
500	64	—	OOM
1,500	4	3.631	fits
1,500	8	7.261	fits
1,500	16	—	OOM

5.1 Fitted memory model

The thirteen successful measurements were fit by ordinary least squares to

$$\widehat{\text{mem}}(N, B) = c_0 + c_1 B + c_2(NB) + c_3(N^2 B) \quad [\text{GB}], \quad (6)$$

whose four terms correspond, respectively, to a fixed constant overhead, a per-batch-element cost independent of N (biases, small dense layers), a cost linear in both N and B (the GRU encoder and fusion/head layers, each processing $N \times B$ independent sequences), and a cost quadratic in N , linear in B (self-attention, Eq. (5), batched). The fit is

$$c_0 = 9.710 \times 10^{-4}, \quad c_1 = 1.615 \times 10^{-5}, \quad (7)$$

$$c_2 = 5.090 \times 10^{-4}, \quad c_3 = 6.400 \times 10^{-8}, \quad (8)$$

with $R^2 = 0.9999999$ across all thirteen points — residuals below 4×10^{-3} GB throughout, i.e. essentially exact given measurement granularity. This fit is the quantitative basis for every projection in the remainder of this note, and for the live calculator at [website/tradinggamescale.html](https://tradinggamescale.html).

5.2 A caveat: peak transient memory exceeds the fitted quantity

Eq. (6) predicts $\widehat{\text{mem}}(1500, 16) \approx 14.5$ GB — comfortably under the 15.57 GB usable ceiling — yet $N=1500, B=16$ was *measured* to fail with an out-of-memory error. `used_memory()` reports resident memory after the step completes; it understates the *peak* memory held transiently during backpropagation through attention, when the softmax output, its gradient, and several intermediate tensors coexist. We do not attempt to model that peak directly; instead, every capacity claim in this note treats the highest value that was *empirically observed to succeed* (8.657 GB, at $N=500, B=32$) as the boundary of a verified-safe region, and treats predictions between that value and the raw usable ceiling as requiring direct verification before being relied on.

6 Proposed Architecture: Anchor-Token Cross-Attention

6.1 Design

Replace Eq. (4)’s all-pairs attention with attention against a small, fixed *context set* $\mathcal{C}$, built from three parts:

- **Anchor stocks.** A fixed subset $\mathcal{A} \subset \{1, \dots, N\}$, $|\mathcal{A}| = K$ (e.g. index or sector leaders). Their embeddings $\{h_j : j \in \mathcal{A}\}$ are already computed as part of the per-candidate GRU pass over all N candidates — selecting them costs $O(1)$, with no additional encoding.
- **Macro token** m_{macro} , from the existing macro-series GRU, kept separate rather than pre-fused with portfolio state.
- **Portfolio token** m_{port} , from the global portfolio scalars, likewise kept separate. Under game v2 the cash token is a further fixed context token ($|\mathcal{C}| = K + 3$); it is not a candidate, so it adds no N -dependent cost.

$$\mathcal{C} = \{h_j : j \in \mathcal{A}\} \cup \{m_{\text{macro}}, m_{\text{port}}\}, \quad |\mathcal{C}| = K + 2. \quad (9)$$

Every candidate, including anchors themselves, queries against this fixed set:

$$z_i = \text{Attn}(Q=h_i, K=V=\mathcal{C}), \quad i = 1, \dots, N. \quad (10)$$

This is precisely the Induced Set Attention Block construction of Lee et al. [2019], with the inducing points replaced by *interpretable, real* anchor stocks and two designated context tokens, rather than learned abstract vectors — a deliberate choice for a system whose outputs should remain auditable (Section 2).

6.2 Why not a single pooled token

An intermediate design — reject in favor of Eq. (9) — would replace the anchor set with a single pooled summary, $m_{\text{pool}} = \frac{1}{N} \sum_i h_i$, and let every candidate attend only to $\{m_{\text{pool}}, m_{\text{macro}}, m_{\text{port}}\}$. This is cheaper still, $O(N \cdot 3)$, but every candidate then reads the *identical* pooled vector: attention over a one-element key set degenerates to a fixed (query-independent, up to normalization) linear combination, discarding exactly the selectivity that motivates attention over mean-pooling in the first place — the same reasoning **StockSwingPredictor** applied when it replaced **DualCNN_v1**’s mean-pool with **DualCNN_v2**’s cross-attention (Section 2). Real, distinguishable anchor stocks let different candidates learn different attention weights over $\mathcal{A}$.

6.3 Complexity

Attention cost is now

$$\text{cost}_{\text{attn}}(N, K) = \Theta(N \cdot (K + 2)), \quad (11)$$

linear in N for fixed K , versus Eq. (5)’s quadratic cost. Figure 1(b) shows the resulting connectivity: outer (candidate) nodes connect only to the small central context set, not to each other.

6.4 Pipeline

Figure 2 shows the full proposed forward pass. The per-candidate GRU encoder, and the actor/critic heads, are unchanged from the baseline; only the aggregation stage between them changes.

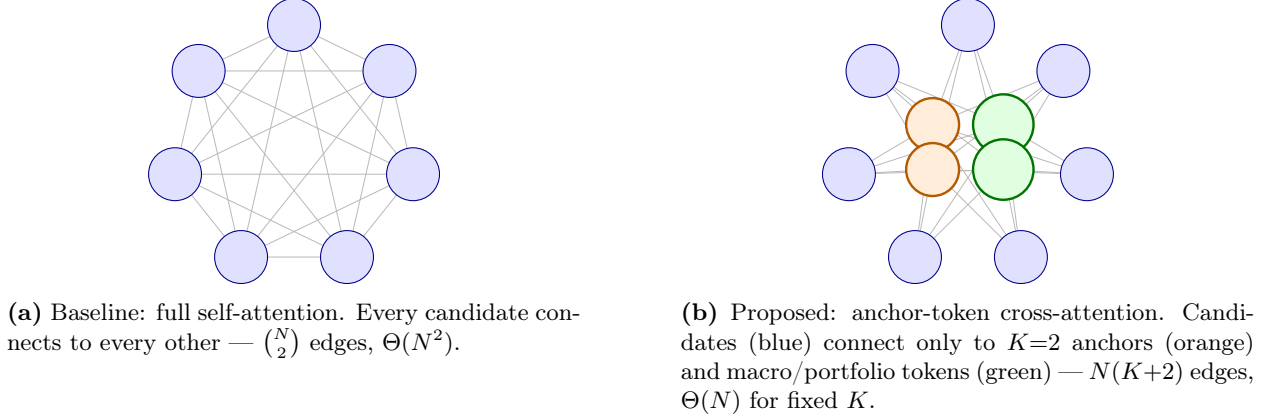


Figure 1: Attention connectivity, illustrated at a small N for legibility.

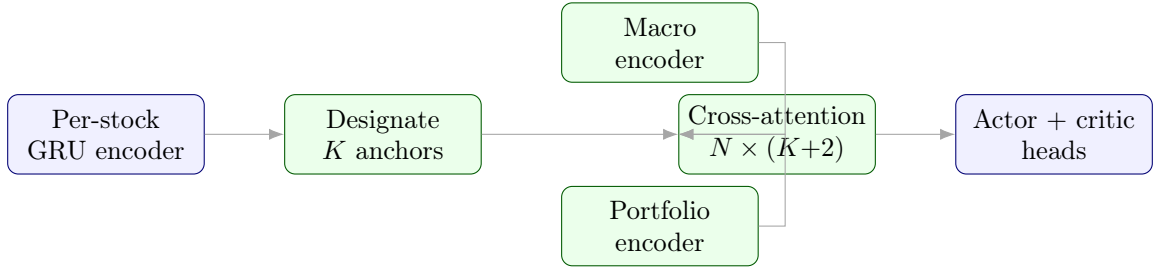


Figure 2: Proposed policy pipeline. Blue: unchanged from the baseline. Green: new or restructured relative to Section 4.

7 The Redesign Is Necessary but Not Sufficient

Substituting Eq. (11) for Eq. (5) inside the fitted model of Eq. (6) — reusing c_0, c_1, c_2 unchanged, since the GRU/fusion/head costs they capture do not depend on the attention pattern, and reusing c_3 as the measured per-key-value-pair cost of attention itself — gives a projected memory function for the proposed design:

$$\widehat{\text{mem}}_{\text{proposed}}(N, K, B) = c_0 + c_1 B + c_2(NB) + c_3(N(K+2)B). \quad (12)$$

As a consistency check, setting $K = N - 2$ (a degenerate anchor set equal to the whole candidate universe) recovers Eq. (6) exactly, as it must: $\widehat{\text{mem}}_{\text{proposed}}(N, N-2, B) = \widehat{\text{mem}}(N, B)$.

Table 2 evaluates both models at $N=1500, K=8$. The proposed design does reduce memory — by roughly 16% at $B=32$ — but *both* figures exceed this GPU’s usable ceiling. The reason is visible directly in Eq. (12): at $N=1500$, the linear term $c_2(NB)$ already dominates the total almost completely ($c_2 \cdot 1500 \cdot 32 \approx 24.4$ GB, versus a quadratic-term contribution, at $K=8$, of only $c_3 \cdot 1500 \cdot 10 \cdot 32 \approx 0.031$ GB). Eq. (11)’s asymptotic improvement is real, but at $N=1500$ it has already stopped being the binding constraint.

Solving Eq. (12) for the largest batch size under a memory budget M ,

$$B_{\max}(N, K, M) = \frac{M - c_0}{c_1 + c_2 N + c_3 N(K+2)}, \quad (13)$$

gives, at $N=1500, K=8$: $B_{\max} \approx 11.4$ under the verified-safe budget ($M=8.657$ GB) and $B_{\max} \approx 20.4$ under the raw usable ceiling ($M=15.57$ GB, subject to the Section 5.2 caveat). Neither approaches the batch sizes (128–256) that fit easily at $N=60$ today. Reaching them at $N=1500$ requires

Table 2: Baseline vs. proposed, projected at $N=1500, K=8$.

	Baseline (GB)	Proposed, $K=8$ (GB)
$B = 4$	3.63	3.06
$B = 8$	7.26	6.12
$B = 32$	29.04	24.46
$B = 256$	232.3	195.7

a second lever independent of the attention redesign — most plausibly a smaller per-candidate embedding width, mixed-precision training, or gradient checkpointing through the GRU — which we leave as future work (Section 9).

8 Training

The policy is trained with Proximal Policy Optimization [Schulman et al., 2017] using Generalized Advantage Estimation [Schulman et al., 2016]; the loop is unaffected by the architecture change proposed here beyond the forward-pass cost analyzed above. Algorithm 1 summarizes the current implementation (`ppo.jl`, `train.jl`).

Algorithm 1 TradingGame training iteration

- 1: Roll out one full episode with the current policy, sampling $a_{i,t}^{\text{type}} \sim \text{Categorical}(\text{softmax}(\ell_{i,t}))$ per candidate; record $(s_t, a_t, \log \pi_{\text{old}}(a_t|s_t), r_t, V(s_t))_{t=1}^T$
 - 2: Compute advantages via GAE: $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, $\hat{A}_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}$; normalize $\hat{A}_t$
 - 3: **for** K_{epochs} passes over shuffled minibatches **do**
 - 4: $\rho_t(\theta) \leftarrow \exp(\log \pi_{\theta}(a_t|s_t) - \log \pi_{\text{old}}(a_t|s_t))$
 - 5: $\mathcal{L}^{\text{CLIP}} \leftarrow \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)]$
 - 6: $\mathcal{L} \leftarrow -\mathcal{L}^{\text{CLIP}} + c_v \mathbb{E}_t[(V_{\theta}(s_t) - \hat{R}_t)^2] - c_e \mathbb{E}_t[\mathcal{H}(\pi_{\theta}(\cdot|s_t))]$
 - 7: Update θ with Adam [Kingma and Ba, 2015]
 - 8: **end for**
 - 9: Checkpoint if the (held-out, greedy) episode return improved
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The continuous buy-weight $w_{i,t}$ (Eq. (2)) is used deterministically — $w_{i,t} = \sigma(\text{logit})$, renormalized by the masking layer — rather than sampled from a second distribution; PPO’s log-probability ratio ρ_t is computed over the categorical a^{type} choice only. This keeps the stochastic side of a hybrid discrete/continuous action space tractable for a hand-rolled implementation, at the cost of not learning position-sizing stochastically. None of this changes under the proposed architecture, which affects only how s_t is encoded into $\ell_{i,t}$ and $V(s_t)$.

9 Discussion and Open Questions

- **Anchor selection.** Candidates include index constituents (e.g. Nifty 50), the top- K candidates by market capitalization, or one representative per major sector. Any choice must be a subset of the episode’s own candidate universe, so no extra encoding pass is needed.
- **Anchor stability.** Fixing $\mathcal{A}$ per episode keeps K genuinely constant for Eq. (11)’s accounting; re-selecting anchors more frequently (e.g. today’s top movers) could be more informative but

reopens a small $O(N)$ selection cost per step and makes the attended set non-stationary within an episode.

- **The linear bottleneck (Section 7)** is, on the evidence here, the harder remaining problem: it is intrinsic to encoding N independent price histories, not to any particular cross-candidate mixing strategy.
- **GPU wiring.** `ppo.jl/train.jl` currently run entirely on CPU (a deliberate scoping decision for the initial PPO implementation); every measurement in this note characterizes the network in isolation, not the full rollout-and-update loop.
- **assemble_observation is a separate, CPU-bound cost** — a per-candidate Julia loop run once per environment step — outside the scope of the GPU analysis here and worth profiling independently as N grows.

10 Conclusion

Full self-attention across the candidate universe, as implemented today, is quadratic in N and — measured directly, not merely argued asymptotically — exhausts a 16.7GB GPU at $N=1500$ between batch sizes 8 and 16. Anchor-token cross-attention, restricted to a small fixed context set of real anchor stocks plus separate macro and portfolio tokens, is linear in N and recovers exactly the selectivity a pooled-summary alternative would have discarded. It reduces memory meaningfully (Table 2) but does not, by itself, make $N=1500$ training cheap: a linear per-candidate encoding cost, fit from the same measurements (Eq. (6)), remains and now dominates. Both findings — the quadratic bottleneck and its linear successor — come from the same thirteen-point empirical characterization and the same closed-form model, which is the artifact we consider most durable here: not a specific proposed architecture, but a validated method for answering "does this fit on this GPU" without guessing.

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